



# Math 4641: Numerical Methods

## Lecture 9 — Spline Interpolation Method

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# Lecture Plan

## The agenda for today

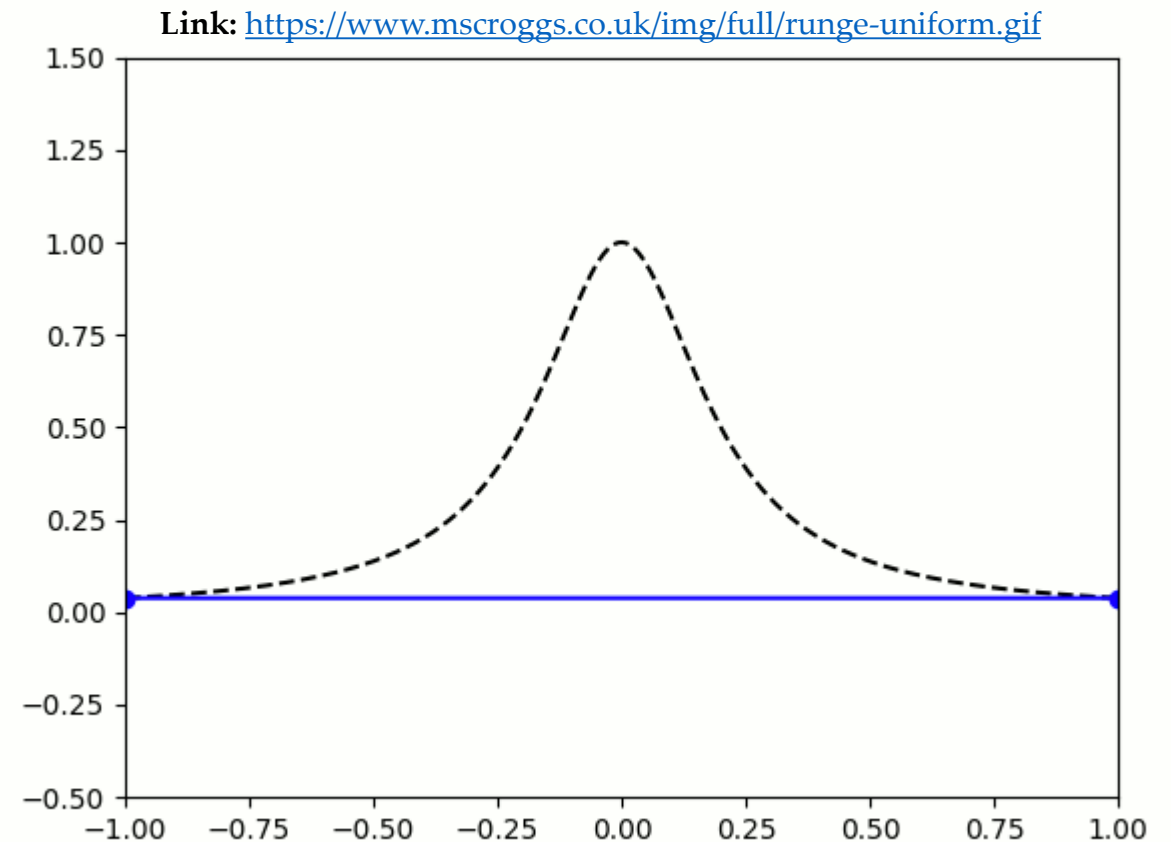
- A small recap of the Runge Phenomenon
- What is Spline Interpolation?
- Understand how Spline Interpolation is immune to the Runge Phenomenon
- Derive the equations for Linear Spline Interpolation
- Derive the equations for Quadratic Spline Interpolation

# Runge Phenomenon

## A small recapitulation

When  $n$  becomes *large*, in many cases, one may get oscillatory behavior in the resulting polynomial. This was shown by Runge when he interpolated data based on a simple function on an interval of  $[-1, 1]$ .

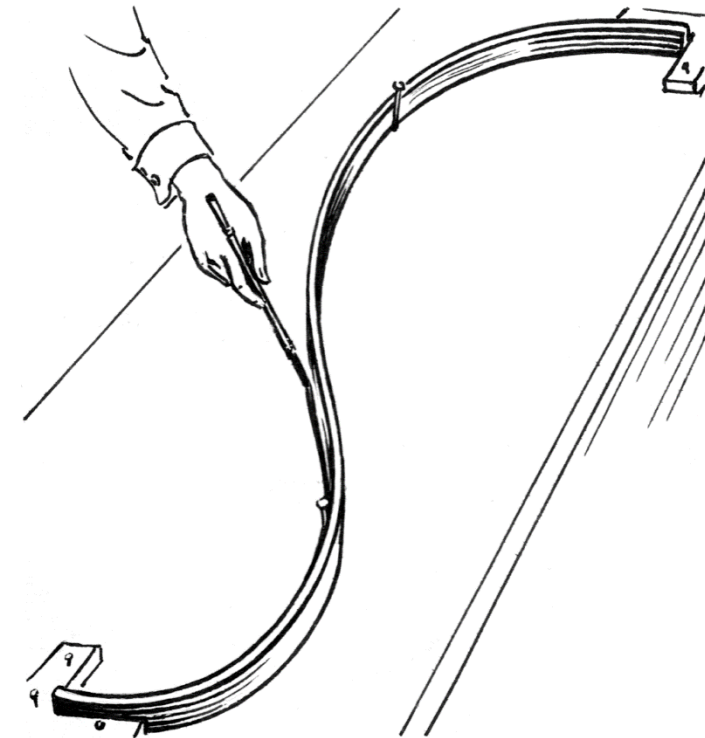
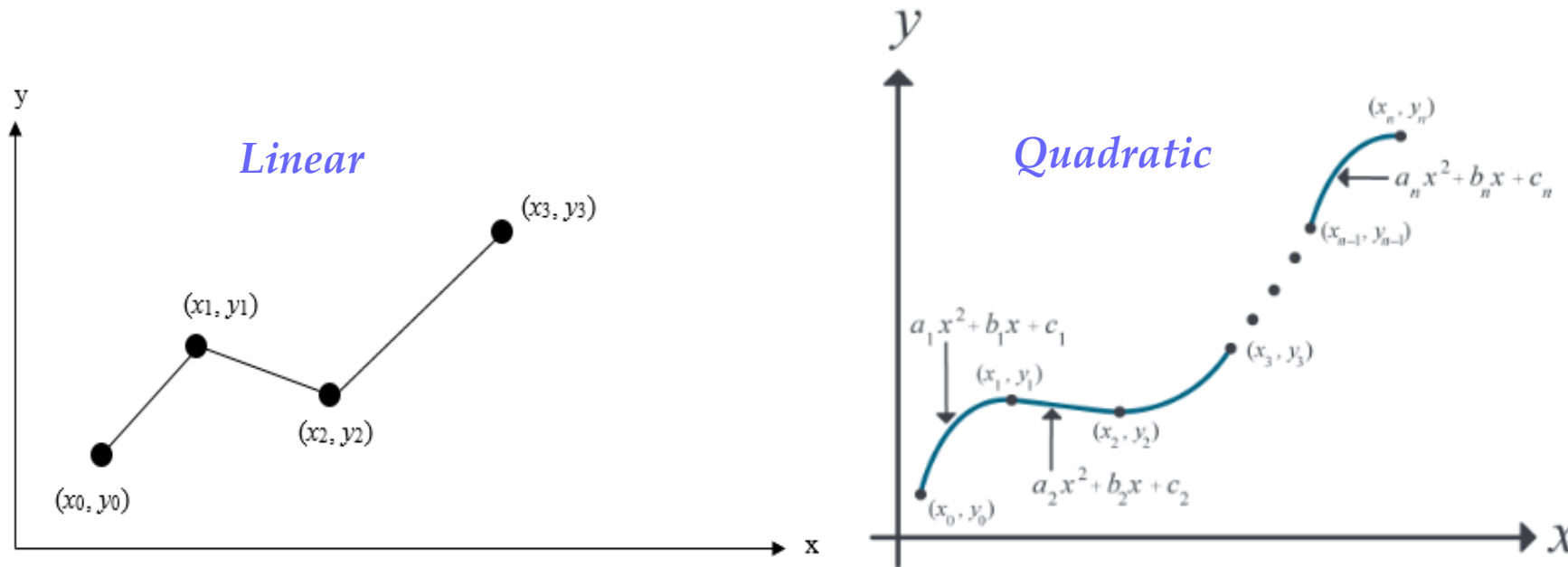
All the interpolation methods that we have covered so far — *Direct* method, *NDD* method, and *Lagrangian* method, all are *susceptible to this phenomenon*.



# Spline Interpolation

What is it?

*Spline* — A function made up of polynomials that each have a specific interval. In other words a "piecewise polynomial function/curve".



*Spline tool*

We can now use information from *more data points*, but at the same time keeping the function *reasonably true to the data behavior*.

# Spline Interpolation

## Linear Spline Interpolation

Assuming that the data is given in ascending order, the interpolating linear spline, also called spline of degree 1,  $f(x)$  is given by

$$\begin{aligned} f(x) &= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0), \quad x_0 \leq x \leq x_1, \\ &= f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1), \quad x_1 \leq x \leq x_2, \\ &\vdots \\ &= f(x_{n-1}) + \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}(x - x_{n-1}), \quad x_{n-1} \leq x \leq x_n. \end{aligned}$$

Note that  $y_i = f(x_i)$  in the above expression and that the terms of

$$\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \quad (2)$$

are simply slopes between  $x_{i-1}$  and  $x_i$ .

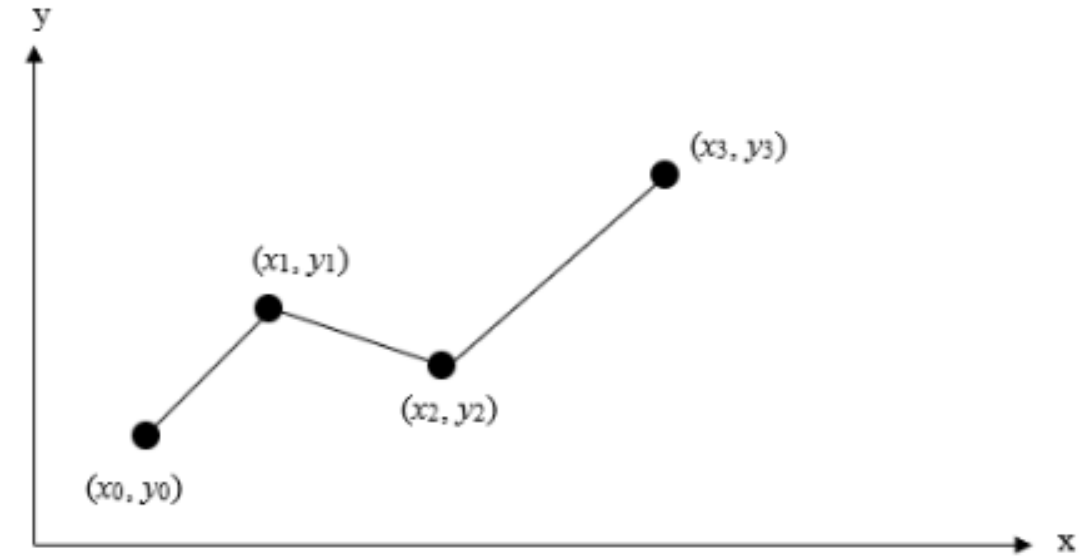


Figure 1. Linear splines.

Remember the equation for a straight line going through 2 points  $(x_1, y_1)$  and  $(x_2, y_2)$ ?

$$\frac{x - x_1}{x_1 - x_2} = \frac{y - y_1}{y_1 - y_2}$$

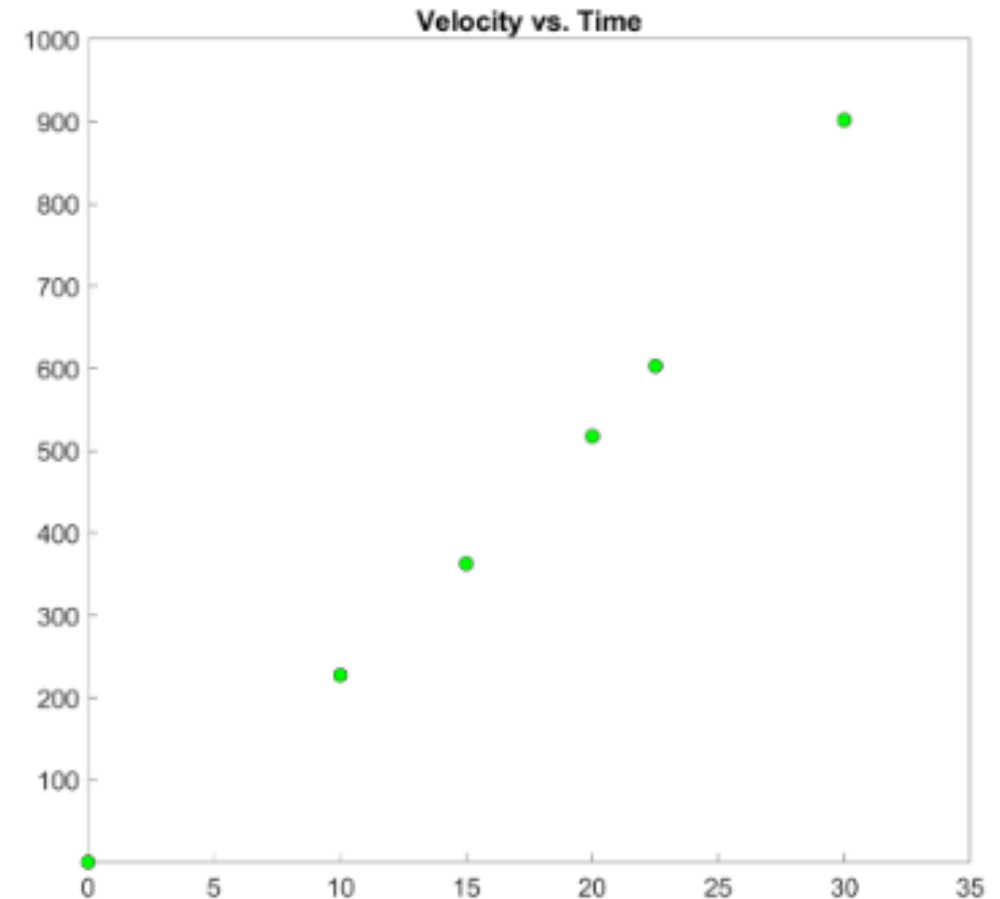
# Spline Interpolation

## A linear spline example

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---------|--------------|
| 0       | 0            |
| 10      | 227.04       |
| 15      | 362.78       |
| 20      | 517.35       |
| 22.5    | 602.97       |
| 30      | 901.67       |



**Figure 2.** Graph of velocity vs. time data for the rocket example.

Determine the value of the velocity at  $t = 16$  seconds using an interpolating linear spline.

# Spline Interpolation

## A linear spline example

### Solution

Since we want to evaluate the velocity at  $t = 16$  and use linear spline interpolation, we need to choose the two data points closest to  $t = 16$  that also bracket  $t = 16$  to evaluate it. The two points are  $t_0 = 15$  and  $t_1 = 20$ .

Then

$$t_0 = 15, v(t_0) = 362.78$$

$$t_1 = 20, v(t_1) = 517.35$$

gives

$$\begin{aligned} v(t) &= v(t_0) + \frac{v(t_1) - v(t_0)}{t_1 - t_0}(t - t_0) \\ &= 362.78 + \frac{517.35 - 362.78}{20 - 15}(t - 15) \\ &= 362.78 + 30.913(t - 15), \quad 15 \leq t \leq 20 \end{aligned}$$

At  $t = 16$ ,

$$\begin{aligned} v(16) &= 362.78 + 30.913(16 - 15) \\ &= 393.7 \text{ m/s} \end{aligned}$$

# Spline Interpolation

## Some caveats about Linear Spline Interpolation

- Linear spline interpolation is *no different* from linear polynomial interpolation.
- It still uses data only from the *two consecutive data points*, and data from other points is not used at all.
- At the interior points of the data, the slope of the spline changes abruptly, which implies that the first derivative is “artificially” *not continuous* at these points.

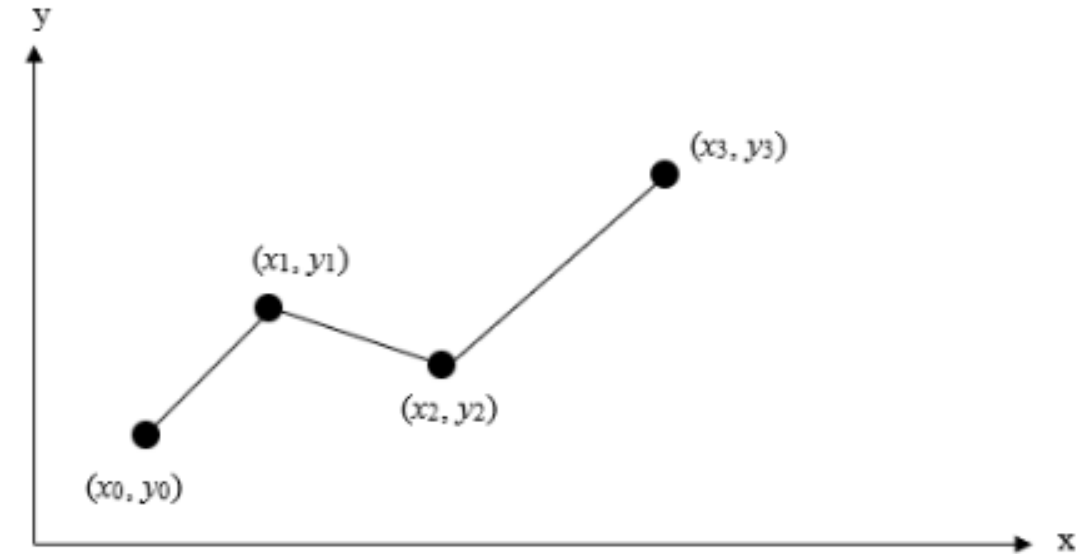


Figure 1. Linear splines.

We need *Quadratic Splines* and above for continuity!



# Spline Interpolation

## Quadratic Spline Interpolation

Quadratic spline interpolation is a method to curve fit data. For quadratic spline interpolation, piecewise quadratics approximates the data between two consecutive data points (Figure 1). Given

$(x_0, y_0), (x_1, y_1), \dots, (x_{n-1}, y_{n-1}), (x_n, y_n)$ , fit an interpolating quadratic spline through the data. The quadratics of the spline are given by

$$\begin{aligned} f(x) &= a_1x^2 + b_1x + c_1, \quad x_0 \leq x \leq x_1 \\ &= a_2x^2 + b_2x + c_2, \quad x_1 \leq x \leq x_2 \\ &\vdots \\ &= a_nx^2 + b_nx + c_n, \quad x_{n-1} \leq x \leq x_n \end{aligned}$$

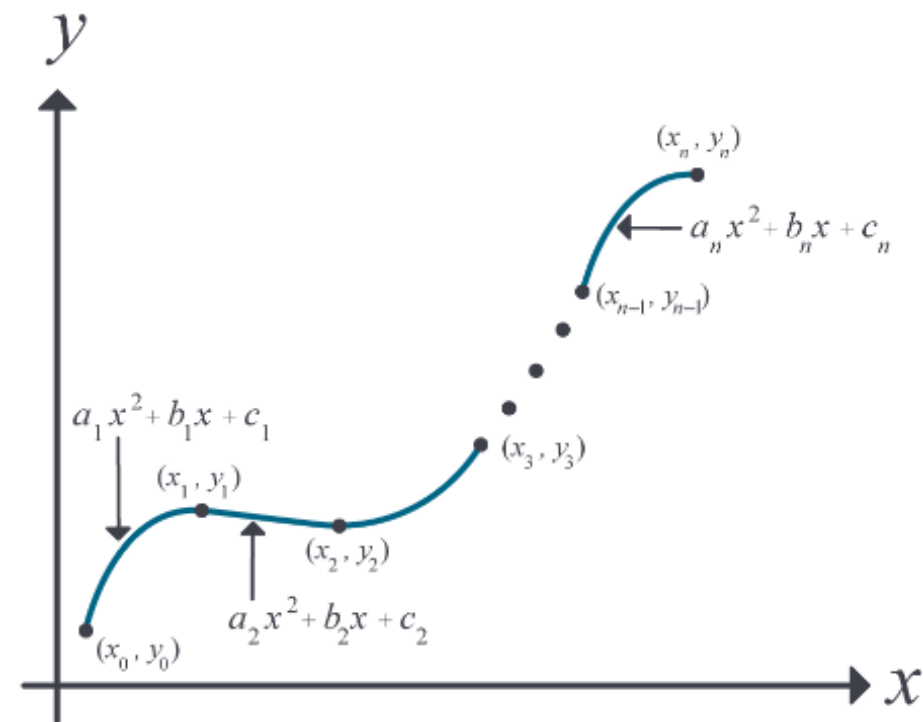
So how does one find the coefficients of these quadratics? There are  $3n$  such coefficients

$$a_i, \quad i = 1, 2, \dots, n$$

$$b_i, \quad i = 1, 2, \dots, n$$

$$c_i, \quad i = 1, 2, \dots, n$$

To find  $3n$  unknowns, one needs to set up  $3n$  equations and then simultaneously solve them. These  $3n$  equations are found as follows.



**Figure 1.** Quadratic spline interpolation

# Quadratic Spline Interpolation

## The first $2n$ equations

1. Each quadratic goes through two consecutive data points

$$a_1x_0^2 + b_1x_0 + c_1 = f(x_0)$$

$$a_1x_1^2 + b_1x_1 + c_1 = f(x_1)$$

$\vdots$

$$a_ix_{i-1}^2 + b_ix_{i-1} + c_i = f(x_{i-1})$$

$$a_ix_i^2 + b_ix_i + c_i = f(x_i)$$

$\vdots$

$$a_nx_{n-1}^2 + b_nx_{n-1} + c_n = f(x_{n-1})$$

$$a_nx_n^2 + b_nx_n + c_n = f(x_n)$$

This condition gives  $2n$  equations as there are  $n$  quadratics going through two consecutive data points.

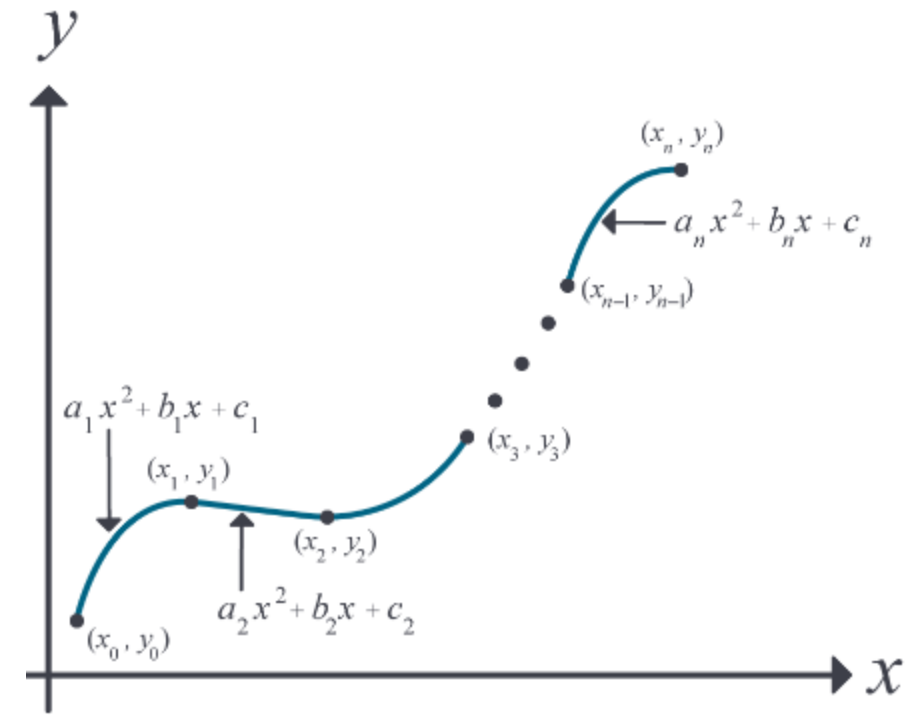


Figure 1. Quadratic spline interpolation

*We need to find  $n$   
more equations!*

# Quadratic Spline Interpolation

## The next $(n - 1)$ equations

2. The first derivatives of two consecutive quadratics are continuous at the common interior points. For example, the derivative of the first quadratic  $a_1x^2 + b_1x + c_1$  is  $2a_1x + b_1$

The derivative of the second quadratic  $a_2x^2 + b_2x + c_2$  is  $2a_2x + b_2$  and the two are equal at the common interior point  $x = x_1$  giving

$$2a_1x_1 + b_1 = 2a_2x_1 + b_2$$

$$2a_1x_1 + b_1 - 2a_2x_1 - b_2 = 0$$

Similarly, at the other interior points,  $x_2, \dots, x_{n-1}$ ,

$$2a_2x_2 + b_2 - 2a_3x_2 - b_3 = 0$$

$\vdots$

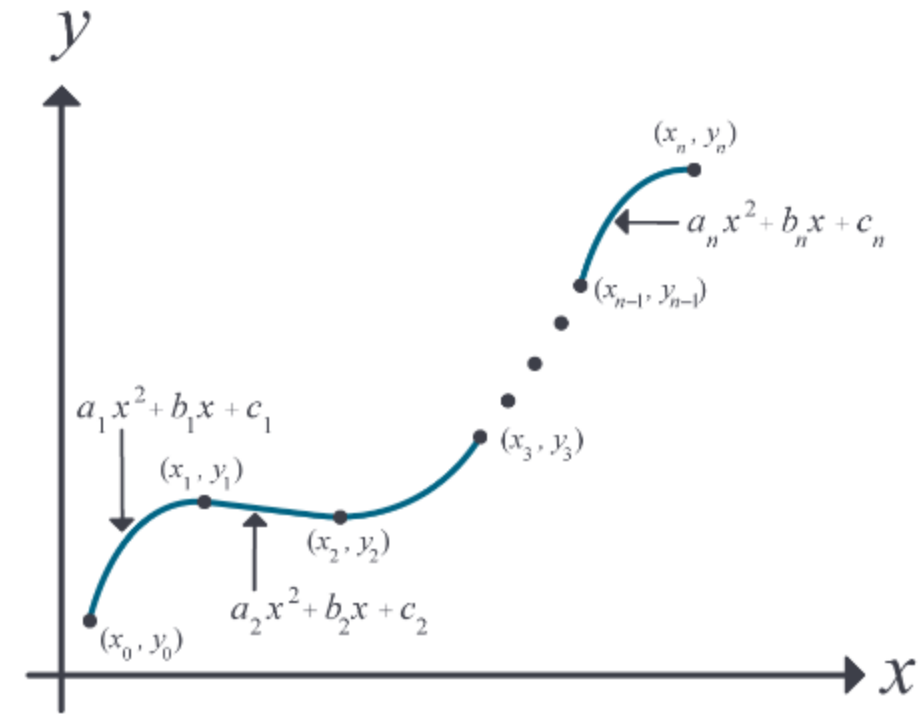
$$2a_ix_i + b_i - 2a_{i+1}x_i - b_{i+1} = 0$$

$\vdots$

$$2a_{n-1}x_{n-1} + b_{n-1} - 2a_nx_{n-1} - b_n = 0$$

For  $(n - 1)$  interior points,  
we get  $(n - 1)$  such  
equations.

*We still need to find  
1 more equation!*



**Figure 1.** Quadratic spline interpolation

# Quadratic Spline Interpolation

## The final equation

So far, the total number of equations obtained is  $2n + (n - 1) = (3n - 1)$  equations.

We still then need one more equation.

We can assume the first quadratic is linear, that is,

$$a_1 = 0$$

Some assume the last quadratic is linear, that is,

$$a_n = 0$$

Others rightly base it on which interval is smaller,  $[x_0, x_1]$  or  $[x_{n-1}, x_n]$ . If  $|x_1 - x_0| \leq |x_n - x_{n-1}|$ , then one would choose  $a_1 = 0$ , else choose  $a_n = 0$ .

This gives us  $3n$  simultaneous linear equations and  $3n$  unknowns. These can be solved by several techniques used to solve a general set of simultaneous linear equations.

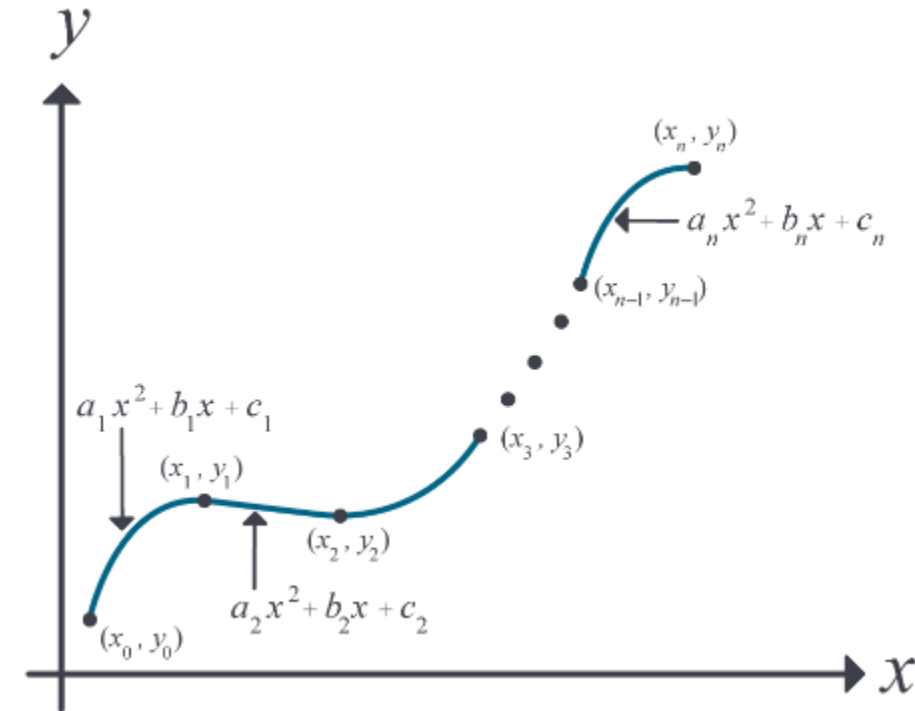


Figure 1. Quadratic spline interpolation

# Mini Quiz

What about Cubic Spline Interpolation?

What is the total number of equations we need for *Cubic Splines*?

How can we obtain them?

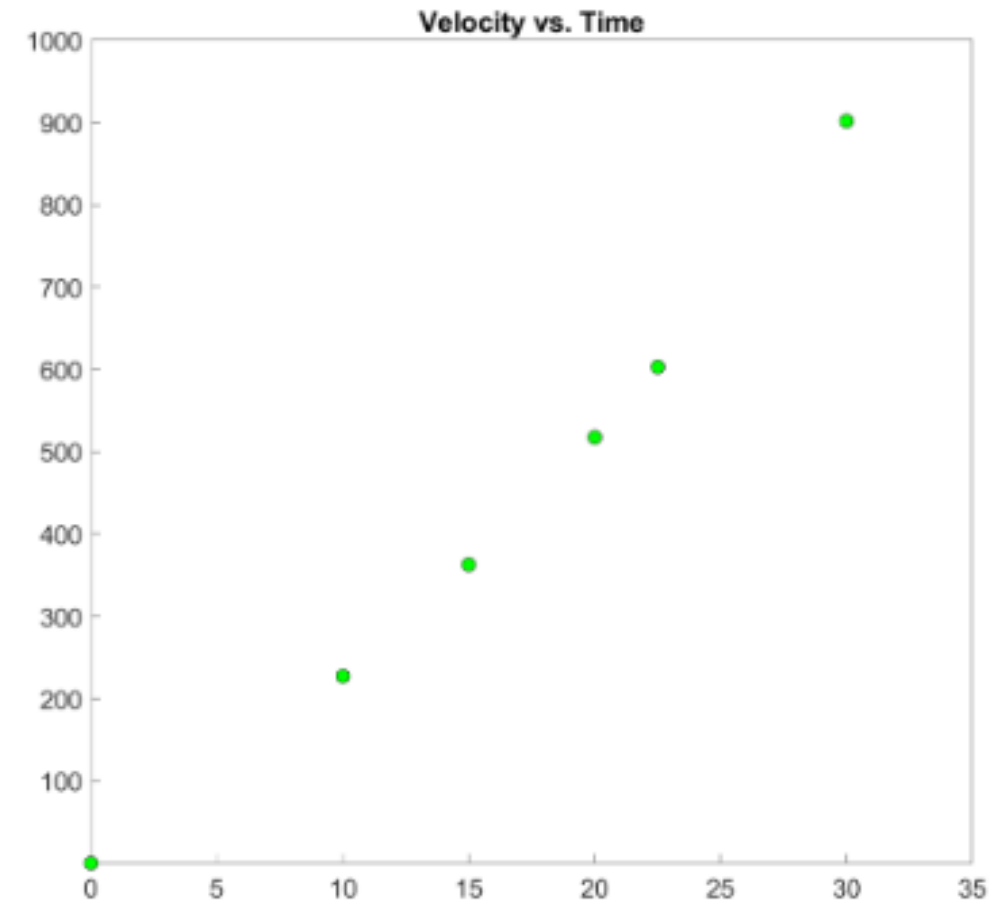
# Spline Interpolation

## A quadratic spline example

The upward velocity of a rocket is given as a function of time in Table 1.

**Table 1.** Velocity as a function of time.

| $t$ (s) | $v(t)$ (m/s) |
|---------|--------------|
| 0       | 0            |
| 10      | 227.04       |
| 15      | 362.78       |
| 20      | 517.35       |
| 22.5    | 602.97       |
| 30      | 901.67       |



**Figure 2.** Graph of velocity vs. time data for the rocket example.

Determine the value of the velocity at  $t = 16$  seconds using quadratic spline interpolation.

# Spline Interpolation

## A quadratic spline example

### Solution

a) Since there are six data points, five quadratics pass through them.

$$\begin{aligned}v(t) &= a_1 t^2 + b_1 t + c_1, \quad 0 \leq t \leq 10 \\&= a_2 t^2 + b_2 t + c_2, \quad 10 \leq t \leq 15 \\&= a_3 t^2 + b_3 t + c_3, \quad 15 \leq t \leq 20 \\&= a_4 t^2 + b_4 t + c_4, \quad 20 \leq t \leq 22.5 \\&= a_5 t^2 + b_5 t + c_5, \quad 22.5 \leq t \leq 30\end{aligned}$$

The equations are found as follows.

1. Each quadratic passes through two consecutive data points.

Quadratic  $a_1 t^2 + b_1 t + c_1$  passes through  $t = 0$  and  $t = 10$ .

$$\begin{aligned}a_1(0)^2 + b_1(0) + c_1 &= 0 \\a_1(10)^2 + b_1(10) + c_1 &= 227.04\end{aligned}$$

Quadratic  $a_2 t^2 + b_2 t + c_2$  passes through  $t = 10$  and  $t = 15$ .

$$\begin{aligned}a_2(10)^2 + b_2(10) + c_2 &= 227.04 \\a_2(15)^2 + b_2(15) + c_2 &= 362.78\end{aligned}$$

Quadratic  $a_3 t^2 + b_3 t + c_3$  passes through  $t = 15$  and  $t = 20$ .

$$\begin{aligned}a_3(15)^2 + b_3(15) + c_3 &= 362.78 \\a_3(20)^2 + b_3(20) + c_3 &= 517.35\end{aligned}$$

Quadratic  $a_4 t^2 + b_4 t + c_4$  passes through  $t = 20$  and  $t = 22.5$ .

$$\begin{aligned}a_4(20)^2 + b_4(20) + c_4 &= 517.35 \\a_4(22.5)^2 + b_4(22.5) + c_4 &= 602.97\end{aligned}$$

Quadratic  $a_5 t^2 + b_5 t + c_5$  passes through  $t = 22.5$  and  $t = 30$ .

$$\begin{aligned}a_5(22.5)^2 + b_5(22.5) + c_5 &= 602.97 \\a_5(30)^2 + b_5(30) + c_5 &= 901.67\end{aligned}$$

# Spline Interpolation

## A quadratic spline example

2. The quadratics have continuous derivatives at the common interior data points.

At  $t = 10$

$$2a_1(10) + b_1 - 2a_2(10) - b_2 = 0$$

At  $t = 15$

$$2a_2(15) + b_2 - 2a_3(15) - b_3 = 0$$

At  $t = 20$

$$2a_3(20) + b_3 - 2a_4(20) - b_4 = 0$$

At  $t = 22.5$

$$2a_4(22.5) + b_4 - 2a_5(22.5) - b_5 = 0$$

3. Assuming the last quadratic  $a_5t^2 + b_5t + c_5$  is linear

$$a_5 = 0$$



# Spline Interpolation

## A quadratic spline example

Think about  
the drawbacks.

$$\begin{bmatrix}
 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 100 & 10 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 225 & 15 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 400 & 20 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 506.25 & 22.5 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 900 & 30 & 1 \\
 20 & 1 & 0 & -20 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 30 & 1 & 0 & -30 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 40 & 1 & 0 & -40 & -1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 45 & 1 & 0 & -45 & -1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0
 \end{bmatrix}
 \begin{bmatrix}
 a_1 \\
 b_1 \\
 c_1 \\
 a_2 \\
 b_2 \\
 c_2 \\
 a_3 \\
 b_3 \\
 c_3 \\
 a_4 \\
 b_4 \\
 c_4 \\
 a_5 \\
 b_5 \\
 c_5
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 227.04 \\
 227.04 \\
 362.78 \\
 362.78 \\
 517.35 \\
 517.35 \\
 602.97 \\
 602.97 \\
 901.67 \\
 0 \\
 0 \\
 0 \\
 0 \\
 0
 \end{bmatrix}$$

# Spline Interpolation

## A quadratic spline example

Solving the above 15 simultaneous linear equations for the 15 unknowns gives

| $i$ | $a_i$    | $b_i$   | $c_i$   |
|-----|----------|---------|---------|
| 1   | -0.15667 | 24.271  | 0       |
| 2   | 1.2021   | -2.9053 | 135.88  |
| 3   | -0.44893 | 46.627  | -235.61 |
| 4   | 2.2315   | -60.589 | 836.55  |
| 5   | 0        | 39.827  | -293.13 |

Therefore, the interpolating quadratic spline is given by

$$\begin{aligned}v(t) &= -0.15667t^2 + 24.271t, \quad 0 \leq t \leq 10 \\&= 1.2021t^2 - 2.9053t + 135.88, \quad 10 \leq t \leq 15 \\&= -0.44893t^2 + 46.627t - 235.61, \quad 15 \leq t \leq 20 \\&= 2.2315t^2 - 60.589t + 836.55, \quad 20 \leq t \leq 22.5 \\&= 39.827t - 293.13, \quad 22.5 \leq t \leq 30\end{aligned}$$

At  $t = 16$  s

$$\begin{aligned}v(16) &= -0.44893(16)^2 + 46.627(16) - 235.61 \\&= 395.50 \text{ m/s}\end{aligned}$$

Not the same  
answer as before!